

Multivariate Calculus*

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1 Continuous Functions

Let ε be a positive real number. In $\mathbb{R}^2$, it is intuitive that a circle of radius ε with center at $\bar{\mathbf{x}}$ is the set of all the points whose distance from $\bar{\mathbf{x}}$ is ε . In $\mathbb{R}^3$ the set of points ε units away from a given point $\bar{\mathbf{x}}$ is a sphere of radius ε centered at $\bar{\mathbf{x}}$. We are going to be interested in the points *inside* a circle or sphere, generalized to $\mathbb{R}^n$, as in the following definition.

Definition 1.1. *The (open) **ball of radius** ε about a point $\bar{\mathbf{x}} \in \mathbb{R}^n$ is the set $\{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - \bar{\mathbf{x}}\| < \varepsilon\}$, also called the ε -**ball about** $\bar{\mathbf{x}}$ and denoted $B(\mathbf{x}, \varepsilon)$ or $B_\varepsilon(\bar{\mathbf{x}})$.*

Note that in $\mathbb{R}^2$ the ε -ball $B(\bar{\mathbf{x}}, \varepsilon)$ is a disc with center at $\bar{\mathbf{x}}$, and in $\mathbb{R}$ the ε -ball is an interval centered at $\bar{\mathbf{x}}$.

Definition 1.2. *A **neighborhood** of a point $\bar{x} \in \mathbb{R}^n$ is a set $\mathcal{N}$ that contains an ε -ball is an interval centered at $\bar{x}$. (In particular, an ε -ball about $\bar{x}$ is itself a neighborhood of $\bar{x}$.)*

Definition 1.3. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function on $\mathbb{R}^n$, let $\bar{\mathbf{x}} \in \mathbb{R}^n$, and let $a \in \mathbb{R}$. If*

$$\forall \varepsilon > 0 : \exists \delta > 0 : \|\mathbf{x} - \bar{\mathbf{x}}\| < \delta \Rightarrow |f(\mathbf{x}) - a| < \varepsilon, \quad (1)$$

we say that a is the limit of f at $\bar{x}$, which we write as $\lim_{x \rightarrow \bar{x}} f(x) = a$. If there is no a in $\mathbb{R}^n$ that satisfies (1) the limit of f at $\bar{x}$ does not exist.

Remark. If $\lim_{x \rightarrow \bar{x}} f(x)$ exists, it is unique.

*This material is largely based on the lecture notes of math camp in Arizona University and Columbia University.

Definition 1.4. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **continuous** at a point $\bar{x} \in \mathbb{R}^n$ if $\lim_{x \rightarrow \bar{x}} f(x) = f(\bar{x})$.

Remark. $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous at $\bar{x}$ if and only if

$$\forall \varepsilon > 0 : \exists \delta > 0 : \|\mathbf{x} - \bar{x}\| < \delta \Rightarrow |f(x) - f(\bar{x})| < \varepsilon$$

i.e., if and only if,

$$\forall \varepsilon > 0 : \exists \delta > 0 : \mathbf{x} \in B(\bar{x}, \delta) \Rightarrow f(\bar{x}) \in B(f(\bar{x}), \varepsilon).$$

Example. Let $\lambda \in \mathbb{R}$ and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = \lambda x$ and let $\bar{x} \in \mathbb{R}$. To verify that f is continuous at $\bar{x}$:

- (i) If $\lambda = 0$, we can simply let δ be any positive number, since $|f(x) - f(\bar{x})| = 0 < \varepsilon$ for any ε .
- (ii) If $\lambda \neq 0$, then for any ε , let $\delta = \frac{1}{|\lambda|}\varepsilon$; then,

$$|x - \bar{x}| < \delta \Leftrightarrow |x - \bar{x}| < \frac{1}{|\lambda|}\varepsilon \Leftrightarrow |\lambda||x - \bar{x}| < \varepsilon \Leftrightarrow |\lambda x - \lambda \bar{x}| < \varepsilon \Leftrightarrow |f(x) - f(\bar{x})| < \varepsilon.$$

Example. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ x - 1, & \text{if } x > 1 \end{cases}$$

Then, f is discontinuous at $\bar{x} = 1$: we have $f(\bar{x}) = 1$, but let $\varepsilon = \frac{1}{2}$, for example. Then, for any $\delta > 0$, there are values of x for which $1 < x < 1 + \delta$ and $0 < f(x) < \frac{1}{2} = 1 - \varepsilon = f(\bar{x}) - \varepsilon$.

Now, we define the continuous functions on metric space. Let (X, d_X) and (Y, d_Y) be metric spaces.

Definition 1.5. Let $\bar{x} \in X$. A function $f : X \rightarrow Y$ is continuous at $\bar{x}$ if for every sequence $\{x_n\}$ that converges to $\bar{x}$, the sequence $\{f(x_n)\}$ converges to $f(\bar{x})$.

Definition 1.6. A function $f : X \rightarrow Y$ is continuous if it is continuous at every point in X .

Theorem 1.7. A function $f : X \rightarrow Y$ is continuous at $\bar{x}$ if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $d_X(x, \bar{x}) < \delta \Rightarrow d_Y(f(x), f(\bar{x})) < \varepsilon$, i.e.,

$$\forall \varepsilon > 0 : \exists \delta > 0 : x \in B(\bar{x}, \delta) \Rightarrow f(x) \in B(f(\bar{x}), \varepsilon).$$

Proof. ($\Rightarrow$): Let $\varepsilon > 0$. Suppose, by way of contradiction, there is no $\delta > 0$ such that $d_X(x, \bar{x}) < \delta \Rightarrow d_Y(f(x), f(\bar{x})) < \varepsilon$, i.e.,

$$\forall \delta > 0 : \exists x \in B(\bar{x}, \delta) \text{ for which } f(x) \notin B(f(\bar{x}), \varepsilon). \quad (2)$$

Then, in particular, for every $n \in \mathbb{N}$, let $\frac{1}{n}$ play the role of δ above: there is an $x_n \in B(\bar{x}, \frac{1}{n})$ for which $f(x_n) \notin B(f(\bar{x}), \varepsilon)$. We therefore have a sequence $\{x_n\}$ in X that converges to $\bar{x}$ but the sequence $\{f(x_n)\}$ does not converge to $f(\bar{x})$, contradicting our assumption that f is continuous.

($\Leftarrow$): Assume that (2) holds and let $\{x_n\}$ be a sequence that converges to $\bar{x}$. In order to show that $\{f(x_n)\}$ converges to $f(\bar{x})$, let $\varepsilon > 0$. According to (2), there is a $\delta > 0$ for which

$$x \in B(\bar{x}, \delta) \Rightarrow f(x) \in B(f(\bar{x}), \varepsilon).$$

Since $\{x_n\} \rightarrow \bar{x}$, we can choose $\bar{n} \in \mathbb{N}$ such that $n > \bar{n} \Rightarrow x_n \in B(\bar{x}, \delta)$. But, then

$$n > \bar{n} \Rightarrow f(x_n) \in B(f(\bar{x}), \varepsilon),$$

i.e., $\{f(x_n)\}$ converges to $f(\bar{x})$ and f is therefore continuous at $\bar{x}$. □

Remark. For functions f from $\mathbb{R}^n$ to $\mathbb{R}^m$, this theorem says that f is continuous at $\bar{x} \in \mathbb{R}^n$ if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\|x - \bar{x}\| < \delta \Rightarrow \|f(x) - f(\bar{x})\| < \varepsilon$.

Definition 1.8. Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function and $S \subset \mathbb{R}^m$ be a set. Then,

$$f^{-1}(S) := \{x \in U \mid f(x) \in S\}$$

Example. Let $f(x) = x^2$, then $f^{-1}([0, 1]) = [-1, 1]$ and $f^{-1}([1, 4]) = [-1, -2] \cup [2, 1]$ and $f^{-1}([-4, -1]) = \emptyset$.

Theorem 1.9. A function $f : X \rightarrow Y$ is continuous if and only if for every open set V in Y the inverse image $f^{-1}(V)$ is an open set in X .

Proof. ($\Rightarrow$): Suppose f is continuous on U and V is open. Let $a \in f^{-1}(V)$ (i.e., $f(a) \in V$). Since V is open, there exists a $\varepsilon > 0$ such that $B(f(a), \varepsilon) \subset V$. Since f is continuous at a , there exists a $\delta > 0$ such that $\|x - a\| < \delta$ implies $\|f(x) - f(a)\| < \varepsilon$.

Then, since $x \in B(a, \delta)$, $f(x) \in B(f(a), \varepsilon)$ and $f(x) \subset V$. Thus, $B(a, \delta) \subset f^{-1}(V)$ and $f^{-1}(V)$ is open.

($\Leftarrow$): Suppose $f^{-1}(V)$ is open when $V \in \mathbb{R}^m$ is open. Given $a \in U$ and $\varepsilon > 0$, $B(f(a), \varepsilon)$ is open. Then, by assumption, $f^{-1}(B(f(a), \varepsilon))$ is open. Since $a \in f^{-1}(B(f(a), \varepsilon))$, there exists a $\delta > 0$ such that $B(a, \delta) \subset f^{-1}(B(f(a), \varepsilon))$. $\square$

Note that the continuous function does not necessarily map open sets to open sets. Above theorem tells us only that the inverse image of an open set under a continuous function is always open. Check below example.

Example. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^2$. When we define the function on an interval, $I = (-1, 1)$. This gives us an image $f(I) = [0, 1)$ and an inverse image $f^{-1}(I) = (-1, 1)$.

Therefore, the image is not open. Indeed, half-open real interval is neither closed nor open set.

Theorem 1.10. *A function $f : X \rightarrow Y$ is continuous if and only if for every closed set S in Y the inverse image $f^{-1}(S)$ is a closed set in X .*

This gives us four equivalent definitions of a continuous function f from X to Y :

- If for every sequence $\{x_n\}$ that converges to $\bar{x}$, the sequence $\{f(x_n)\}$ converges to $f(\bar{x})$.
- If for every $\bar{x} \in X : \forall \varepsilon > 0 : \exists \delta > 0 : x \in B(\bar{x}, \delta) \Rightarrow f(x) \in B(f(\bar{x}), \varepsilon)$.
- If the inverse image of any open set in Y is an open set in X .
- If the inverse image of any closed set in Y is a closed set in X .

Remark. When the target space Y is actually a normed vector space, it is natural to define the sum and scalar multiple of continuous functions pointwise - i.e., the functions $f + g : X \rightarrow Y$ and $\alpha f : X \rightarrow Y$ are defined by $\forall x \in X : (f + g)(x) = f(x) + g(x)$ and $\forall x \in X : (\alpha f)(x) = \alpha f(x)$. Then, the set $C(X, Y)$ of all continuous functions on X into Y , with these definitions of addition and scalar multiplication, is a vector space.

Exercise 1. *Prove the above remark. This requires showing that $C(X, Y)$ is “closed under vector addition and scalar multiplication”. This does not mean that $C(X, Y)$ is a closed set, but rather that if f and g are in $C(X, Y)$ and $\alpha \in \mathbb{R}$, then $f + g$ and αf are in $C(X, Y)$, i.e., that the sum of continuous functions is a continuous functions and that a multiple of a continuous function is a continuous function.*

For real-valued functions (i.e., if $Y = \mathbb{R}$), we can also define the product fg and (if $\forall x \in X : f(x) \neq 0$) the reciprocal $1/f$ of functions pointwise, and we can show that if f and g are continuous then so are fg and $1/f$.

Remark. If X, Y , and Z are metric spaces and if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then the composition $f \circ g : X \rightarrow Z$ is continuous.

In Euclidean space (i.e., $\mathbb{R}^n$ with any norm) we say that a set is **compact** if it is both closed and bounded. One of the most important properties of continuous functions is that they “preserve” compactness, i.e., if X is a compact subset of $\mathbb{R}^n$ and if $f : X \rightarrow \mathbb{R}^m$ is a continuous function, then the image of X , $f(X)$, is a compact set in $\mathbb{R}^m$. This is the Weierstrass Theorem. In fact, the Weierstrass Theorem holds in general metric spaces:

Theorem 1.11 (Weierstrass Theorem). *If X is compact and $f : X \rightarrow Y$ is continuous, then $f(X)$ is a compact subset of Y .*

Corollary 1.11.1. *If $f : X \rightarrow \mathbb{R}$ is a continuous real-valued function on a compact set, then f attains a maximum and a minimum on X .*

Corollary 1.11.1 tells us that every continuous function whose domain is a compact set C actually achieves its global max and its global min on C . This is the result one needs to prove that equilibria do exist in a wide range of economic models.

2 Derivatives

2.1 Derivatives of single-variate functions

Definition 2.1. *Let $A \subset \mathbb{R}$ and $x \in A$. A function $f : A \rightarrow \mathbb{R}$ is said to be **differentiable at x** if the limit*

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. In that case, we define the derivative of f at x as the limit above and denoted as $f'(x)$.

Definition 2.2. *We say a function $f : A \rightarrow \mathbb{R}$ is said to be **differentiable** if f is differentiable at any $x \in A$.*

Clearly, if a function f is differentiable at x , then it is continuous at x as:

$$\begin{aligned} \lim_{x' \rightarrow x} [f(x') - f(x)] &= \lim_{x' \rightarrow x} \left[\frac{f(x') - f(x)}{x' - x} (x' - x) \right] \\ &= \lim_{x' \rightarrow x} \left[\frac{f(x') - f(x)}{x' - x} \right] \lim_{x' \rightarrow x} (x' - x) \\ &= f'(x) \cdot 0 = 0. \end{aligned}$$

However, a function continuous at x may fail to be differentiable at x , since the function may have a kink point.

Derivatives of some commonly used functions:

- $(x^\alpha)' = \alpha x^{\alpha-1}$
- $(\ln x)' = 1/x$
- $(e^x)' = e^x$
- $(\sin x)' = \cos x$

Since the derivative is essentially the limit of the function, it inherits the properties of functions.

- $(f + g)'(x) = f'(x) + g'(x)$.
- $(\lambda f)' = \lambda f'$.
- $(fg)' = f'g + fg'$.
- $(f/g)' = \frac{f'g - fg'}{g^2}$.

Note that if a function f is differentiable at point x , then f is continuous at x . This is because the differentiability is defined on limit. However, the converse is not true. Check the following counterexamples.

Example. $f(x) = |x|$ is not differentiable at $x = 0$.

Proof.

$$f(0^-) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \frac{-x}{x} = -1, \text{ and}$$

$$f(0^+) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \frac{x}{x} = 1$$

Since there are two limit points, the differentiability is not defined on $x = 0$. □

2.2 Total derivatives

Now we generalize the notion of derivatives to multivariate functions.

Definition 2.3. Let $A \subset \mathbb{R}^n$ and $x \in A$. A function $f : A \rightarrow \mathbb{R}^m$ is said to be **differentiable at x** if there exists an $m \times n$ real matrix C such that

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Ch\|}{\|h\|} = 0$$

In this case, define the **(total) derivative of f at x** as the matrix C , denoted as $f'(x)$ or $Df(x)$.

Definition 2.4. A function $f : A \rightarrow \mathbb{R}$ is said to be **differentiable** if A is open and f is differentiable at any $x \in A$ and denote the derivative function of f as f' .

Extending total derivative to a function such as $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by applying the definition of derivative similarly, we have the total derivative of F about $\mathbf{x}^*$ as:

$$F(\mathbf{x}^* + \Delta \mathbf{x}) - F(\mathbf{x}^*) \approx \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix}$$

where $\mathbf{x}^* = (x_1^*, \dots, x_n^*)$. The matrix in the equation is often called as **Jacobian** of F at $\mathbf{x}^*$.

2.3 Partial derivatives

Definition 2.5. Let $A \subset \mathbb{R}^n$ and $x \in A$. For a function $f : A \rightarrow \mathbb{R}^m$, its **partial derivative of the i -th coordinate w.r.t. the j -th argument at $x \in A$** is

$$\frac{\partial f_i}{\partial x_j}(x) = \left. \frac{d}{dt} f_i(x + te_j) \right|_{t=0}$$

where the vector e_j is the j -th canonical basis of $\mathbb{R}^n$.

Example. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

By definition of partial derivatives, we have

$$\begin{aligned} \frac{\partial f}{\partial x}(0, 0) &= \left. \frac{d}{dt} f(t, 0) \right|_{t=0} = \lim_{t \rightarrow 0} \frac{f(t, 0) - f(0, 0)}{t - 0} \\ &= \lim_{t \rightarrow 0} \frac{0 - 0}{t - 0} = 0 \end{aligned}$$

and

$$\begin{aligned}\frac{\partial f}{\partial y}(0,0) &= \left. \frac{d}{dt} f(0,t) \right|_{t=0} = \lim_{t \rightarrow 0} \frac{f(0,t) - f(0,0)}{t - 0} \\ &= \lim_{t \rightarrow 0} \frac{0 - 0}{t - 0} = 0\end{aligned}$$

so, both of the partial derivatives of f exist. However, f is not differentiable at $(0,0)$. In fact, f is not even continuous at $(0,0)$. To see this, notice that f constantly take the value $1/2$ along the path $y = x^2$ except for at the point $(0,0)$ where the f takes the value 0 .

As in above example, the existence of partial derivative does not imply differentiability of f at x . However, if for each (i,j) , the partial derivative, $\frac{\partial f_i}{\partial x_j}(x)$ exists not only at x , but also on an open ball around x , and $\frac{\partial f_i}{\partial x_j}(x)$ is continuous at x , then f is differentiable at x .

2.4 Chain Rule

Definition 2.6. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **continuously differentiable** (or C^1) on an open set $U \subset \mathbb{R}^n$ if and only if for each i , $(\partial f / \partial x_i)(\mathbf{x})$ exists for all $\mathbf{x}$ in U and is continuous in $\mathbf{x}$.

This definition can be extended to higher-order derivatives. If $f'(x)$ has a derivative $f''(x)$ for all $\mathbf{x}$ in U and f'' is continuous, we say that f is **twice continuously differentiable** or denote as C^2 on U . Similarly, if all the derivatives of f of order up to k exist and are continuous on U , we say f is k -times continuously differentiable on U or denote as C^k .

Proposition 2.7 (Chain Rule). Let S be a subset of $\mathbb{R}$ and $f : S \rightarrow \mathbb{R}$. Let T be a set such that $f(S) \subset T \subset \mathbb{R}$, and $g : T \rightarrow \mathbb{R}$. If f is differentiable at x , and g is differentiable at $f(x)$, and we have

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$

Above proposition can be generalized to a vector space, where $f : S \rightarrow \mathbb{R}^m$ and $g : T \rightarrow \mathbb{R}^k$. Now, the matrix multiplication is involved. Because $g'(f(x))$ is an $k \times m$ matrix and $f'(x)$ is an $m \times n$ matrix, their product $g'(f(x))f'(x)$ is $k \times n$ matrix, which is exactly the size $(g \circ f)'(x)$ should have.

Exercise 2. Find the first derivative of following functions using Chain rule: $f(x) = \exp(ax)\sin(bx)$.

$$\begin{aligned}f'(x) &= a \exp(ax) \sin(bx) \\ &\quad + b \exp(ax) \cos(bx)\end{aligned}$$

2.5 L'Hospital's Rule

Theorem 2.8. Suppose f and g are real and differentiable in (a, b) and $g'(x) \neq 0, \forall x \in (a, b)$, $-\infty \leq a < b \leq +\infty$. Suppose $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = A$. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = A$$

Exercise 3. Prove $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$.

↗ rewrite $\left(1 + \frac{1}{x}\right)^x = \exp\left(\ln\left(\frac{x+1}{x}\right)^x\right) = \exp\left(x \ln\left(\frac{x+1}{x}\right)\right)$

3 Approximations

3.1 Mean Value Theorem

Theorem 3.1. Let f be defined on $[a, b]$; if f has a local maximum (minimum) at a point $x \in (a, b)$ and if $f'(x)$ exists, then $f'(x) = 0$.

$= \exp\left(\lim_{x \rightarrow \infty} \frac{\ln\left(\frac{x+1}{x}\right)}{\frac{1}{x}}\right)$

using L'Hospital $= \exp\left(\lim_{x \rightarrow \infty} \frac{x}{x+1}\right) = e$.

Note that the converse of above theorem is not true. For example, for a function $f(x) = x^3$, $f'(0) = 0$, although $x = 0$ is neither a minimum nor a maximum.

Theorem 3.2 (Rolle's Theorem). $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and C^1 on (a, b) . If $f(a) = f(b) = 0$, then there is a point $c \in (a, b)$ such that $f'(c) = 0$.

Proof. If f is constant on $[a, b]$, then $f(a) = f(b) = f(c) = 0$, thus $f'(c) = 0$. If f is not constant, f is sometimes positive on (a, b) . By Weierstrass Theorem, f achieves maximum at some point, $c \in [a, b]$. Since $f(c) > 0$, it must lie in the interval (a, b) and $f'(c) = 0$. $\square$

Using the Rolle's theorem, we can prove mean value theorem. Graphically, Mean Value Theorem tells us that if one draws a line segment between any two points on f (which is C^1 function), there exists a point between two points that is parallel to the line segment we drew.

Theorem 3.3 (Mean Value Theorem). Let $f : U \rightarrow \mathbb{R}$ be a C^1 function on U in $\mathbb{R}$. For any points $a, b \in U$, there exists a point $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a) \Leftrightarrow \frac{f(b) - f(a)}{b - a} = f'(c)$$

Proof. Construct a function, $g_0(x) \equiv f(b) - f(x) + \frac{f(b)-f(a)}{b-a}(x-a)$. Then, we have $g_0(a) = 0$ and $g_0(b) = 0$. Applying Rolle's Theorem, we know there exists a $c \in (a, b)$ such that $g'_0(c) = 0$. Thus, plugging $x = c$ in

$$g'_0(x) = -f'(x) + \frac{f(b) - f(a)}{b - a},$$

we have Mean Value Theorem.

Exercise 4. Prove that $\frac{x}{1+x} < \ln(a+x) < x$, for $x > 0$.

Handwritten notes:
 Let $f(x) = \ln x$.
 $\Rightarrow \frac{f(1+x) - f(1)}{(1+x) - 1} = \frac{1}{x}$ $\square$
 $\Rightarrow \ln(1+x) < x$
 $\Rightarrow \frac{1}{1+x} < \frac{1}{x}$
 $\Rightarrow \frac{1}{1+x} < \frac{1}{x} \Rightarrow \frac{1}{1+x} < \frac{1}{x} \Rightarrow \frac{1}{1+x} < \frac{1}{x}$

3.2 Taylor's Theorem

We could also define a best quadratic approximation to a function f , or to the function $F(\Delta x) = f(\bar{x} + \Delta x) - f(\bar{x})$. Here we are going to go further and define a best approximation of order n , the best n -th degree polynomial approximation of F , for any $n \in \mathbb{N}$. This theorem is important because a lot of economics functions are nonlinear and can only be solved by approximating using Taylor polynomials.

Definition 3.4. If f has a derivative f' on an interval and f' is itself differentiable, we denote the derivative of f' by f'' and call f'' the second derivative of f . Continuing in this manner, we obtain functions,

$$f, f', f'', f^{(3)}, \dots, f^{(n)}$$

each of which is the derivative of the preceding one. $f^{(n)}$ is called the n -th derivative or the derivative of order n of f .

Theorem 3.5 (Taylor's Theorem). Suppose f is a real function on $[a, b]$, n is a positive integer, $f^{(n-1)}$ is continuous on $[a, b]$, $f^{(n)}$ exists for every $t \in (a, b)$. Let α, β be distinct points of $[a, b]$ and define,

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k$$

Then, there exists a point x between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n.$$

Note that Taylor's Theorem when $n = 1$ just degenerates the Mean Value Theorem. In general, the theorem shows that f can be approximated by a polynomial of degree $n - 1$, and allows us to estimate the error, if we know bounds on $|f^{(n)}(x)|$.

Example. Let $f(t) = \exp(t)$ and $\alpha = 0$ and $\beta = x$. we have

$$\exp(x) = 1 + x + \frac{1}{2}x^2 + \cdots + \frac{1}{n!}x^n + \frac{\exp(s)}{(n+1)!}x^{n+1}$$

where s is some number between 0 and x .

Since

$$\lim_{x \rightarrow 0} \frac{\frac{\exp(s)}{(n+1)!}x^{n+1}}{x^n} = 0,$$

we can also write,

$$\exp(x) = 1 + x + \frac{1}{2}x^2 + \cdots + \frac{1}{n!}x^n + o(x^n), \quad (x \rightarrow 0)$$

4 Implicit Function Theorem

So far, we have seen the explicit functions. **Explicit functions** are the ones where the endogenous variable is an explicit function of the endogenous variables as $y = f(x_1, \dots, x_n)$. In this explicit form, the endogenous variable y is separated from $\mathbf{x}$, therefore, we call them as an explicit function. On the other hand, **implicit function** does not separate the variables as $G(x_1, x_2, \dots, x_n, y) = 0$. We can think of an example such as:

Example. $y^2 - 5xy + 4x^2 = 0$.

Example. A profit function: $\Pi(x) = pf(x) - wx$, where the prices p and w is given as exogenous variables and $f(x)$ is endogenous variable.

Theorem 4.1 (Implicit Function Theorem). *Let $G(x, y) = c$ be a continuously differentiable function on a ball about (x_0, y_0) in $\mathbb{R}^2$. Suppose that $G(x_0, y_0) = c$ and consider $G(x, y) = c$. If $\partial G(x_0, y_0)/\partial y \neq 0$, then there exists a function $y = y(x)$ that is differentiable on I about the point x_0 such that $G(x, y(x)) \equiv c$ for all x in I , $y(x_0) = y_0$ and*

$$y'(x_0) = -\frac{\partial G(x_0, y_0)/\partial x}{\partial G(x_0, y_0)/\partial y}$$

The implicit function theorem tells us that if there is a function $y(x)$ that solves $G(x, y) = c$ and if it is differentiable, then as x changes by Δx , y will change by $y'(x_0) \cdot \Delta x$. By

(around (x_0, y_0))

applying the Chain rule, we can see how the implicit function theorem appear. Differentiating $G(x, y(x)) = c$ by the Chain rule with respect to x at the point x_0 yields:

$$\frac{\partial G}{\partial x}(x_0, y_0) + \frac{\partial G}{\partial y}(x_0, y(x_0))y'(x_0) = 0.$$

Of course, solving for $y'(x_0)$, the existence of $G(x, y) = c$ needs to be guaranteed and differentiable.

Now, let's extend the result of Implicit function theorem for a function that has range of $\mathbb{R}^k$. For a function f from $A \subset \mathbb{R}^n \times \mathbb{R}^m$ to $\mathbb{R}^k$ and a point $(x_0, y_0) \in A$, the **Jacobian matrix** $D_x f(x_0, y_0)$ at (x_0, y_0) is a $k \times n$ matrix defined as the derivative of $f(x, y_0)$ viewed as a function of x , evaluated at $x = x_0$. Similarly, the Jacobian matrix $D_y f(x_0, y_0)$ at (x_0, y_0) is a $k \times m$ matrix defined as the derivative of $f(x_0, y)$ viewed as a function of y , evaluated at $y = y_0$.

Theorem 4.2. *Let f be a function from $A \subset \mathbb{R}^n \times \mathbb{R}^m$ to $\mathbb{R}^m$. Let $(x_0, y_0) \in \text{int}(A)$ such that $f(x_0, y_0) = 0$. If f is C^1 at (x_0, y_0) and the $m \times m$ Jacobian matrix $D_y f(x_0, y_0)$ is invertible, then there exists an open ball B_x around x_0 and an open ball B_y around y_0 such that $\forall x \in B_x$, there exists a unique $y \in B_y$ such that $f(x, y) = 0$. Therefore, the equation $f(x, y) = 0$ implicitly defines a function $g : B_x \rightarrow B_y$ with the property*

$$f(x, g(x)) = 0$$

for any $x \in B_x$. Furthermore, we know that the function g is differentiable at any $x \in B_x$, and

$$D_x g(x) = -[D_y f(x, g(x))]^{-1} D_x f(x, g(x))$$

Proof. Suppose that there exists an open ball B_x around x_0 such that $\forall x \in B_x$ there exists a unique $y \in B_y$ such that $f(x, y) = 0$. Then, we define $g : B_x \rightarrow B_y$ as $g(x) := y$ such that $f(x, y) = 0$. Then, we know that $f(x, g(x)) = 0$ for any $x \in B_x$. Suppose that g is differentiable at any $x \in B_x$. Then, think of both sides of the equation $f(x, g(x)) = 0$ as a function in x and take derivative, and we should have

$$\frac{d}{dx} f(x, g(x)) = \frac{d}{dx} 0$$

Clearly, the right-hand side of the equation is 0. Consider the function $h : B_x \rightarrow B_x \times B_y$, defined as $h(x) = (x, g(x))$, for any $x \in B_x$. Then, we know that $f(x, g(x)) = f(h(x))$, and

therefore,

$$\begin{aligned}\frac{D}{D_x}f(x, g(x)) &= D_x f(x, g(x))I_n + D_y f(x, g(x))D_x g(x) \\ &= D_x f(x, g(x)) + D_y f(x, g(x))D_x g(x)\end{aligned}$$

Therefore, we have

$$D_y f(x, g(x))D_x g(x) = -D_x f(x, g(x))$$

Since $D_y f(x_0, y_0) = D_y f(x_0, g(x_0))$ is invertible, we have $\det(D_y f(x_0, g(x_0))) \neq 0$. It can be shown that $\det(D_y f(x, g(x)))$ is continuous in x , and therefore we set B_x to be small enough, such that $\det(D_y f(x, g(x))) \neq 0$ for any $x \in B_x$. Therefore, the matrix $D_y f(x, g(x))$ is invertible for any $x \in B_x$ and we finally have

$$D_x g(x) = -[D_y f(x, g(x))]^{-1} D_x f(x, g(x))$$

□

The implicit function theorem can be used often in comparative statistics problem. Suppose $\phi : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$, which we will write as $\phi(x; q)$. The decision maker gets to choose x to maximize ϕ , while q are given parameters over which the decision maker has no control. Sometimes x is referred to as the endogenous variable and q as the exogenous variable. The basic comparative statistics question is how the decision maker will adjust her optimal choice $x^*(q)$ in response to changes in the parameters q . In general, there can be multiple maximizers $x^*(q)$, but we will assume that there exists a unique solution here.

Suppose $\phi(x; q)$ is a smooth objective function, where x refers to n -dimensional vector of choice (endogenous) variables and q refers to the m dimensional vector of exogenous parameters. Consider the simple maximization problem: $x^*(q) = \arg \max \phi(x; q)$. Suppose ϕ is strictly concave and differentiable, so the first order condition defines the maximizer: $f(x; q) = D_x \phi(x; q) = 0_n^T$.

The Jacobian of f is $D_x f = D_{xx} \phi(x; q)$ and is nonsingular since ϕ is strictly convex. The implicit function theorem says we can treat $x^*(q)$ as continuously differentiable function, at least locally. Moreover, we can compute the local response of the $x^*(q)$ to small changes in q :

$$\begin{aligned}D_q x^*(\bar{q}) &= -[D_{xx} f(x^*(\bar{q}); \bar{q})]^{-1} D_{qx} f(x^*(\bar{q}); \bar{q}) \\ &= -[D_{xx} \phi(x^*(\bar{q}); \bar{q})]^{-1} D_{qx} \phi(x^*(\bar{q}); \bar{q})\end{aligned}$$

By the chain rule, the change of the value function $\phi^*(q) = \phi(x^*(q); q)$ in response to changes in q is

$$\begin{aligned} D_q \phi(x^*(\bar{q}); \bar{q}) &= D_q \phi(x, q)|_{x=x^*(\bar{q}), q=\bar{q}} + D_x \phi(x, q) D_q x^*(q) \\ &= D_q \phi(x, q)|_{x=x^*(\bar{q}), q=\bar{q}} \end{aligned}$$

In other words, for small changes, the “second order effect ” of how the maximizer x^* responds to q is irrelevant; simply compute the “first order effect” of how q changes the objective function evaluated at the fixed maximizer $x^*(q)$. This observation is sometimes called the “Envelope Theorem” (which we will discuss more in detail in Optimization section).

If x and q are scalars, $D_q x^*(q)$ becomes

$$\frac{\partial x^*}{\partial q} = \frac{\partial^2 \phi / \partial q \partial x}{\partial^2 \phi / \partial x^2}$$